NOTATION · CHEAT SHEET

Distribution Notation at a Glance

The marks from the notation cards across the distribution pages: how each is read and what it means. Every group links to the full card on its page.

DISTRIBUTIONS

$(\Omega, \mathcal{F}, \mathbb{P})$

A PROBABILITY SPACE
Outcomes, events, and the probability measure.

$\mathbb{P}_X(A)$

THE DISTRIBUTION OF X
Equals $\mathbb{P}(X \in A)$.

$X \overset{d}{=} Y$

EQUAL IN DISTRIBUTION
Same law, not necessarily the same values.

BINOMIAL

$X \sim \mathrm{Bin}(n, p)$

BINOMIAL, N TRIALS, P
The number of successes in n independent trials.

$q = 1 - p$

Q, THE FAILURE PROBABILITY
An abbreviation, not a new parameter.

GEOMETRIC

$\mathrm{Geom}(p)$

GEOMETRIC WITH P
Two conventions: count trials, or count failures.

$q^{k-1}p$ vs $q^k p$

THE TWO PMFS
The exponent shows which convention is in use.

NEGATIVE BINOMIAL

$\mathrm{NB}(r, p)$

NEGATIVE BINOMIAL
Trials until the r-th success.

$k = r, r{+}1, \ldots$

THE SUPPORT STARTS AT R
Fewer trials cannot hold r successes.

HYPERGEOMETRIC

$\mathrm{Hyp}(N, K, n)$

HYPERGEOMETRIC
Population N with K successes; n draws without replacement.

$$\frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}$$

THE PMF
Successes chosen, failures chosen, over all samples.

POISSON

$\mathrm{Pois}(\lambda)$

POISSON WITH RATE LAMBDA
Counts of events in a window.

λ

LAMBDA
The expected count per window; it depends on the window size.

DISCRETE UNIFORM

$\mathrm{DU}(a, b)$

DISCRETE UNIFORM
Each integer from a to b equally likely.

$\frac{1}{b-a+1}$

ONE OVER THE COUNT
The $+1$ counts both endpoints.

CONTINUOUS UNIFORM

$X \sim U(a, b)$

UNIFORM FROM A TO B
The slots are the endpoints, not a centre and a spread.

$\frac{1}{b-a}$

ONE OVER THE WIDTH
A flat density; no $+1$ here.

EXPONENTIAL

$\mathrm{Exp}(\lambda)$

EXPONENTIAL WITH RATE LAMBDA
Some books put the scale $\beta = \frac{1}{\lambda}$ in the slot.

$P(X > x) = e^{-\lambda x}$

THE SURVIVAL TAIL
The chance of waiting longer than x.

NORMAL

$N(\mu, \sigma^2)$

NORMAL, MEAN MU
Second slot: usually the variance; some texts use σ.

$Z \sim N(0, 1)$

CAPITAL Z, STANDARD NORMAL
A score $z = \frac{x-\mu}{\sigma}$ counts standard deviations.

$\varphi(z),\ \Phi(z)$

PHI; CAPITAL PHI
The standard normal density and its CDF.

SAME MARKS, DIFFERENT JOB

$N(\mu, \sigma^2)$	Check whether the second slot is the variance or the standard deviation.
$\mathrm{Exp}(\lambda)$	Rate or scale: the two readings give means $\frac{1}{\lambda}$ and λ.
$U(a, b)$ vs $\mathrm{DU}(a, b)$	Width $b - a$ against a count of $b - a + 1$ values.
Φ vs φ	Capital is the CDF, lowercase the density.